Design Document: ViralIQ AI Analysis Engine

1. Executive Summary

ViralIQ is an advanced image analysis platform designed to predict the "viral potential" of Instagram posts. It leverages a hybrid approach combining **Computer Vision (CV)** for technical objectivity and **Large Language Models (LLMs)** for subjective aesthetic analysis.

2. Technical Stack

Layer	Technology	Purpose
Frontend	React.js	User interface and result visualization
Backend	Node.js / Express	API orchestration and file management
CV Engine	Python / OpenCV / Pillow	Technical image analysis (metrics, ratios)
AI Engine	Groq API (qwen/qwen3.8-27b)	Subjective aesthetic and viral assessment
Storage	Local File System	Temporary image staging via Multer

---

3. Backend Architecture & Theory

The backend operates as an orchestration layer that triggers two independent analysis pipelines in parallel to ensure maximum efficiency and comprehensive scoring.

A. Technical Analysis (Python/OpenCV)

The system spawns a Python process to execute `analyzer.py`. This layer focuses on **objective data**:

- **Technical Quality:** Analysis of blur, noise, and resolution.
- **Aspect Ratio Compatibility:** Checking if the image fits Instagram's preferred ratios (e.g., 4:5 for portraits) to avoid unwanted cropping.
- **Metrics:** Extraction of raw image data (dimensions, format, etc.).

B. Subjective Analysis (Groq Vision AI)

The image is converted to Base64 and sent to the Groq Vision API with a specialized prompt. The AI acts as a professional curator, scoring four key qualitative dimensions:

1. **Background Aesthetics:** Evaluation of lighting, color harmony, and clutter.
2. **Foreground Clarity:** Assessment of subject prominence and focus.
3. **Composition:** Checking for framing, visual hierarchy, and the Rule of Thirds.
4. **Emotional Appeal:** Evaluating originality, mood, and likely audience resonance.

---

4. Scoring Algorithm & Weightage

To arrive at the final **Overall Viral Score**, ViralIQ uses a weighted average formula. This ensures that while technical quality is important, the "vibe" and emotional appeal (which drive virality) carry more weight.

The Formula

$$\text{Overall Score} = (T \times 0.15) + (R \times 0.15) + (B \times 0.15) + (F \times 0.15) + (C \times 0.20) + (A \times 0.20)$$

Terms Defined:

- T **(Technical Quality):** 15% — Basic image clarity.
- R **(Aspect Ratio):** 15% — Platform compatibility.
- B **(Background Aesthetics):** 15% — Visual cleanliness.
- F **(Foreground Clarity):** 15% — Subject focus.
- C **(Composition):** 20% — Artistic framing.
- A **(Engagement Potential):** 20% — Emotional/Viral hook.

Scoring Tiers

The final calculated score is categorized into a "Viral Tier" to give the user an immediate qualitative verdict:

Score Range	Tier Label	Interpretation
$81 - 100$	**Algorithm's Favourite**	Exceptional quality; high probability of viral reach.
$61 - 80$	**High Potential**	Strong image; likely to perform well with the right caption.
$41 - 60$	**Promising**	Average quality; needs specific improvements to stand out.
$0 - 40$	**Low Potential**	Poor technicals or aesthetics; high risk of being ignored.

---

5. Frontend Integration

Data Flow

1. **Input:** User uploads an image via the React frontend → sent as `multipart/form-data` to the `/analyze` endpoint.
2. **Processing:** The backend runs the Python and Groq pipelines → calculates the weighted score → generates a JSON response.
3. **Output:** The frontend receives the JSON and maps it to:
    - A **Circular Gauge** for the Overall Score.
    - A **Breakdown List** showing the 6 individual scores.
    - A **Text-based Analysis** showing the AI's Verdict, Strengths, Weaknesses, and Actionable Suggestions.

API Connection

The frontend connects to `http://localhost:5000/analyze` using an asynchronous POST request. The connection is seamless, allowing the user to receive a professional-grade image audit in seconds.